

A Comparative Study of Transfer Learning Models for Weed and Crop Classification Using Deep Learning

Srisabari Prahadeshwar G
Department of CSE
JAIN (Deemed-to-be-University)
Bengaluru, India
srisabari499@gmail.com

Dr. Suriya Prakash J
Department of CSE
JAIN (Deemed-to-be-University)
Bengaluru, India
Suriyaprakash.j@jainuniversity.ac.in

Mikhail Sai
Department of CSE
JAIN (Deemed-to-be-University)
Bengaluru, India
mikhils25@gmail.com

Gaduputi Kotlo Vijaya
Department of CSE
JAIN (Deemed-to-be-University)
Bengaluru, India
vijayagaduputikotlo@gmail.com

Jaivishanth N C
Department of CSE
JAIN (Deemed-to-be-University)
Bengaluru, India
jaivishanthchandraskar5656@gmail.com

Pritesh Jain
Department of CSE
JAIN (Deemed-to-be-University)
Bengaluru, India

Abstract- Weeds that aren't controlled are still one of the biggest threats to sustainable food production. They not only lower crop yields, but they also make herbicides more expensive, which harms soil health and local ecosystems in ways that last for a long time. It is slow, unreliable, and impossible to do by hand at the scale that modern farming needs to do it. This study examines the efficacy of three lightweight, ImageNet-pretrained architectures MobileNetV1, MobileNetV2, and EfficientNetB0 in automating tasks applied to the V2 Plant Seedlings Dataset, encompassing twelve distinct plant classes. Instead of treating each backbone like a black box, we connect them all to a single classification head made up of Global Average Pooling, Batch Normalisation, and Dropout layers. We then train all of the models in the same two-phase way: 70:20:10 across training, validation, and test partitions, and a suite of augmentation operations is used throughout the training, validation, and test partitions to make the models more robust in real-world situations. For all three networks, we give information on accuracy, precision, recall, F1-score, and confusion-matrix breakdowns. EfficientNetB0 was the clear winner, with a test accuracy of 98.05. MobileNetV1 (87.79) and MobileNetV2 (81.42) were far behind. The results show that transfer learning is a useful and effective way to classify agricultural images. They also give developers who are making weed-management tools for precision farming contexts a clear performance standard to aim for.

Index Terms- Transfer Learning, Deep Learning, Weed Classification, Convolutional Neural Networks, MobileNetV1, MobileNetV2, EfficientNetB0, Precision Agriculture, Plant Seedling Classification, Image Classification, Data Augmentation, Fine-Tuning

I. INTRODUCTION

To feed a planet whose population keeps growing, every hectare of farmland needs to work at or near its best. Weeds contribute to the process through competition between mature plants for soil moisture, sun rays, minerals, and root space, thereby reducing the harvest by two-figure percentages. There

are also disadvantages in adopting conventional approaches such as weeding teams and herbicides applied over large areas. Inaccurate spraying of chemicals speeds up the resistance of weeds and negatively impacts soil bacteria; moreover, the nearby water sources might get contaminated [1]. As such, it is essential to have a tool capable of distinguishing between weeds and crop plants in real-time.

Deep learning, and more specifically, CNN, has revolutionized plant detection based on images. CNNs create their representations layer by layer, starting with low-level edge detectors and moving up to high-level shape descriptors. They do this without using hand-crafted feature pipelines. The practical problem is that training a deep network from a random start needs both a lot of labelled data and a lot of GPU time, which agricultural research groups don't always have. Transfer learning elegantly avoids this problem: a model that has already been trained on ImageNet has a lot of information about textures, contours, and structural patterns that works well with plant images with much less data and computing power [2].

Numerous research teams have validated the potential of this methodology. Espejo-Garcia et al. [3] demonstrated that finetuned convolutional networks consistently outperform handcrafted feature methods on real-world crop-weed images, while Vypirailenko et al. [2] showed that ResNet, MobileNet, and DenseNet backbones all generalise meaningfully to weed growth-stage classification. What the literature still lacks, however, is a carefully controlled head-to-head comparison of the three lightweight models MobileNetV1, MobileNetV2, and EfficientNetB0 evaluated on the same twelve-class seedling benchmark under identical experimental conditions.

This paper fills that gap. We put all three architectures through the same preprocessing pipeline, augmentation strat-

egy, and two-phase training protocol. Then we compare their performance using the same set of metrics. The goal isn't just to pick a winner; it's also to learn about the trade-offs between accuracy and efficiency that people have to make when choosing a backbone for a field robot or edge device.

II. LITERATURE SURVEY

The interaction between deep learning and precision agriculture has attracted consistent research attention over the past decade. One of the recurring themes is the problem of identifying weeds from crops at the seedling stage, where the features like shape, color, and growth rate become almost indistinguishable to an untrained eye or even classical machine vision approaches [5].

Jose et al. [1] tackled this problem using transfer learning on a heterogeneous data of crops and weeds, showing that CNNs pre-trained on other tasks performed much better than classifiers trained from scratch. These results were among the first to provide a consistent benchmark for this problem specifically. Vypirailenko et al. [2] considered the aspect of weed growth stage separately, proving that learned features contain information about the changes in phenology that cannot be detected with hand-crafted representations. Espejo-Garcia et al. [3] took the next step, combining fine-tuned backbones like MobileNet, DenseNet, and Xception with traditional classifiers, proving that deep features could act as a good front-end to classifiers, even when the final classification was done by SVM or random forest.

These findings have been corroborated by more extensive studies. Hasan et al. [5] produced an inventory of the unsolved problems in weed detection, noting that variations in lighting conditions, cluttered environments, imbalanced classes, and the visual similarity among species remain the principal sources of errors. Mujtaba et al. [4] put this problem into perspective regarding the larger context of agricultural AI, arguing that accurate classification is crucial for real-world applications, such as autonomous spraying bots and drone surveillance systems. Kamilaris and Prenafeta-Boldu [8] similarly observed that contemporary deep learning architectures generally surpass manually constructed pipelines once a sufficiently diverse training set is accessible.

With respect to the architecture, the depthwise separable convolution trick was first proposed by Howard et al. [12], which made the first version of MobileNet fast enough to perform inference on-device. Later, Sandler et al. [13] enhanced this approach with inverted residual structures and linear bottleneck layers to make MobileNetV2 more efficient. In contrast, Tan and Le [11] chose a completely different approach to develop EfficientNet, where they applied an exponentiated compound coefficient to scale network depth, width, and input size. As we can observe from the above discussion, for weed classification, both accuracy and efficiency trade-offs are involved.

Even so, none of these studies compares MobileNetV1, MobileNetV2, and EfficientNetB0 models using the twelve class plant seedlings V2 dataset within the same experimental

framework. The table below provides a brief summary of the works that are most directly related to the current study.

TABLE I
SUMMARY OF KEY RELATED WORKS

Ref.	Core Contribution	Relevance to This Work
[1]	Weed/crop classification via transfer learning	Direct methodological pre-decessor
[2]	DL for weed growth-stage detection	Validates learned-feature effectiveness
[3]	Transfer learning for weed ID assistance	Closely aligned objective
[4]	ML/DL solutions across agriculture	Situates work in broader AI context
[5]	Survey of DL weed-detection methods	Defines open challenges
[12], [13]	MobileNetV1 and MobileNetV2	Justifies lightweight backbone choice
[11]	EfficientNet compound scaling	Strongest accuracy reference model

III. PROPOSED SYSTEM

A. Dataset and Preprocessing

The proposed experiments make use of the V2 Plant Seedlings Dataset, an open-source dataset comprising images of 12 different classes of plants taken when they were seedlings. These classes include economically important crops, as well as weed types that might commonly be found in agriculture. As can be seen from Fig. 1, the distribution of the classes is slightly unbalanced, with Common Chickweed being the most prevalent and Cleavers being the least prevalent.

All images are resized to the required size input by each of the backbones and then normalized with the mean and standard deviations of the channels computed over ImageNet data. In addition, all images used for training are also augmented randomly through flipping images both horizontally and vertically, rotating images up to 30°, and applying zoom variations. By doing so, different augmentations make the networks concentrate on learning discriminative plant structures rather than any irrelevant factors such as framing and angles of illumination of the images. The dataset is divided into training, validation, and testing sets with proportions of 70%, 20%, and 10% respectively.

B. Model Architectures

Three transfer-learning backbones are evaluated:

MobileNetV1 [12] employs depthwise-separable convolutions instead of ordinary convolutions, resulting in significantly lower parameters and adequate representational power. The backbone's scale factors can be configured based on specific deployment requirements.

MobileNetV2 [13] enhances the concept of depthwise separability through the use of inverted residual blocks that utilize linear bottleneck structures. The use of a bottleneck is essential for ensuring that any information that could otherwise be lost in dimensionality reduction is retained.

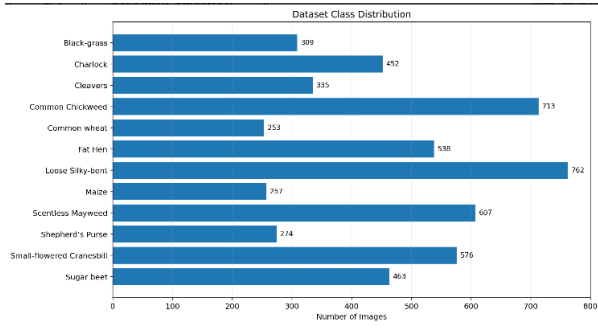


Fig. 1: Distribution of images across the twelve plant classes in the V2 Plant Seedlings Dataset.

EfficientNetB0 [11] follows a more principled path when it comes to scaling by tuning not just the depth, but the width, and even the input resolution, by means of a single compound factor. Indeed, even in its most modest form, B0, the design manages to strike an appealing trade-off between accuracy and efficiency.

The backbones are initialized using the weights obtained from pretraining on ImageNet, and the pretrained weights in the base layers are fixed for the network to learn the new head in the new domain.

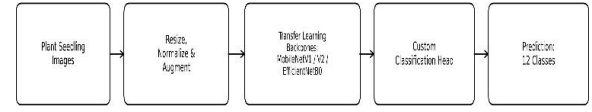
C. Classification Head and Training Protocol

One standard classifying head is added to the top of each backbone so that all models can be compared on the same level. The classifying head is made up of one Global Average Pooling layer (that makes each spatial dimension collapse into one activation vector for each channel), a Batch Normalization layer (that normalizes the activations and serves as a regularization technique), a Dropout layer (that randomly zeros out some activations during training), and a fully connected softmax output layer with twelve neurons. The training takes place in two phases. In **Phase 1**, the weights of the backbone are frozen while the weights of the classification head are learned during a predetermined number of epochs, which enables the classification head to fine-tune the feature space of the backbone without disrupting the weights in the process. In **Phase 2**, the whole model is unfrozen, and learning happens in an end-to-end fashion using a considerably lower learning rate, which enables the backbone to adjust itself to distinguish between the fine differences in appearance of the plant seedlings.

Algorithm III-C summarizes the complete pipeline.

- 1: Load the V2 Plant Seedlings Dataset
- 2: Resize images and apply channel normalization
- 3: Apply randomized augmentation (flip, rotate, zoom, shift)
- 4: Partition dataset: 70% train / 20% val / 10% test
- 5: Initialize MobileNetV1, MobileNetV2, EfficientNetB0 with ImageNet weights
- 6: Attach shared classification head to each backbone
- 7: **Phase 1:** Freeze backbone; train classification head

Project Workflow



Training → Validation → Fine-Tuning → Evaluation (Accuracy, Loss, Confusion Matrix)

Fig. 2: End-to-end workflow of the proposed weed classification system.

- 8: **Phase 2:** Unfreeze all layers; fine-tune at reduced learning rate
- 9: Evaluate on held-out test set; record accuracy and loss
- 10: Generate confusion matrices; compile comparison table

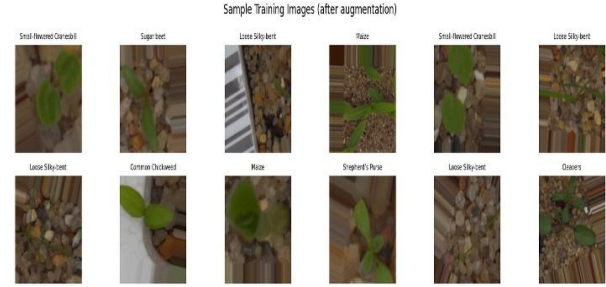


Fig. 3: Sample images from the Plant Seedlings Dataset showing different classes.

The process of the design process starts with the collection and pre-processing of data, followed by augmentation and division of datasets. After which, the data would be provided to the transfer learning algorithms where the features are extracted and classified. These models are assessed with test data, and parameters like accuracy and losses are computed. Lastly, comparisons between models would be made to find the most optimal algorithm.

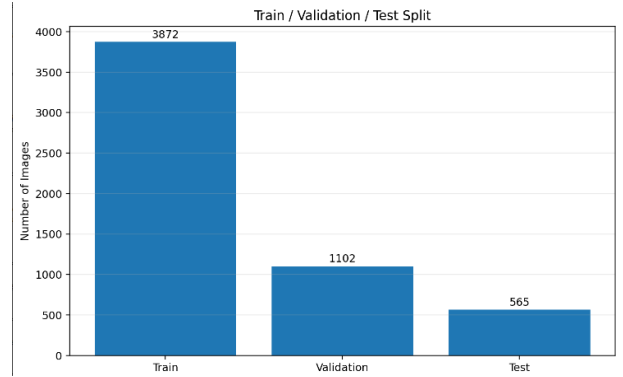


Fig. 4: Training, validation, and testing split.

IV. RESULTS AND DISCUSSION

A. Training Dynamics

Analyzing the trends for all three neural networks, there is a significant difference in the way they behave almost immediately after starting the training process. EfficientNetB0 shows the best trends regarding convergence in terms of learning curve stability, where its accuracy increases consistently until reaching a high peak point with minimal fluctuation, and the validation loss drops gradually to reach a low minimum point. However, MobileNetV1 falls somewhere in between. Its accuracy curve increases in much the same pattern, but tapers off quite considerably compared to EfficientNetB0, indicating that the depth separable convolutions, which lie at the heart of MobileNetV1 architecture, manage to learn discriminative features that are adequate but far from perfect for this particular problem. MobileNetV2 performs rather poorly: although an improved version of MobileNetV1, it produces more variable curves on validation, terminating at a significantly lower level of accuracy. An intuitive reason behind such performance can be found in the nature of the inverted residual blocks used within the network.

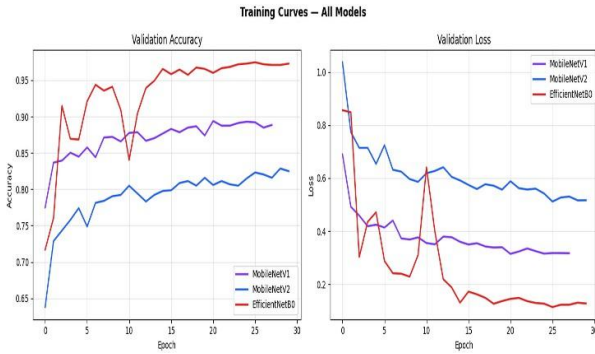


Fig. 5: Training and validation accuracy/loss curves for all three models across epochs.

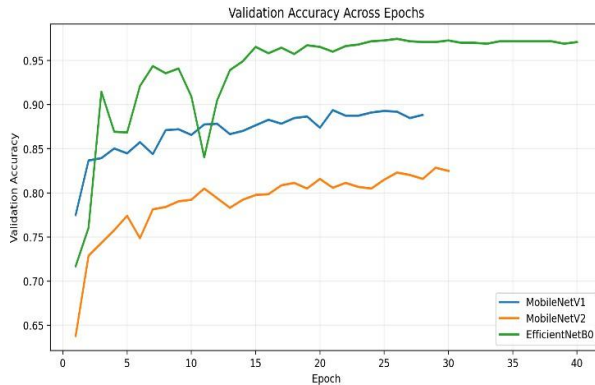


Fig. 6: Validation accuracy trajectories for MobileNetV1, MobileNetV2, and EfficientNetB0.

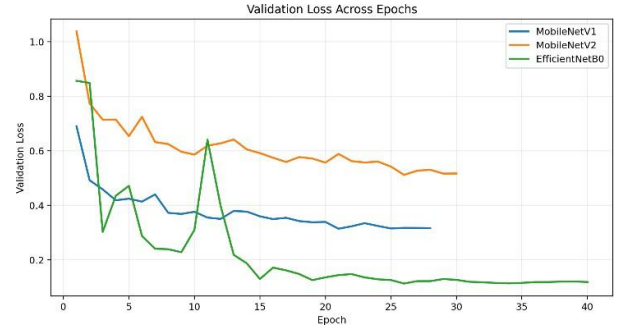


Fig. 7: Validation loss trajectories for MobileNetV1, MobileNetV2, and EfficientNetB0.

B. Test Set Performance

EfficientNetB0 obtains an impressive 98.05% accuracy on the test data split, while the loss stands at 0.0868; an excellent result considering the twelve-class classification challenge wherein some of the classes look almost similar. On its part, MobileNetV1 attains an accuracy of 87.79% with a loss value of 0.3422, while MobileNetV2 obtains 81.42% and 0.5843 loss (Fig. 8, Fig. 9).

A difference of around 16% between the two models is significant enough to cause practical challenges in a deployment environment. An automated sprayer which cannot accurately identify weeds 18% of the time would end up destroying the crop through applying excessive amounts of weed killer or leaving patches of weeds unattended both situations which cost money. Therefore, EfficientNetB0's superior accuracy translates to improved system performance.

That being said, it would be unfair to write off the other models since they have their strengths. Their compact nature and reduced inference latency means they perform better in embedded systems with limited computational capabilities. In such a case, choosing the appropriate backbone model would depend on the specific use case. If accuracy is critical, EfficientNetB0 is preferred, but in hardware-limited environments, MobileNetV1 is the way to go.

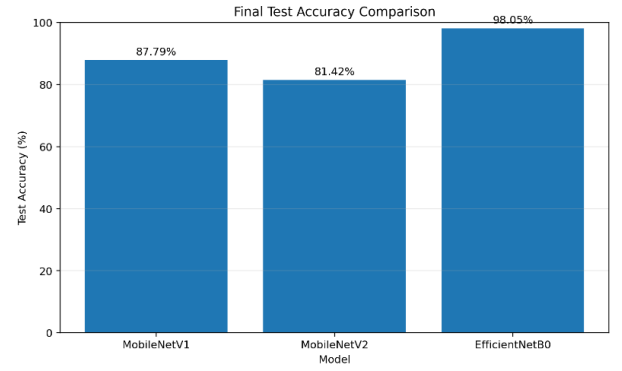


Fig. 8: Final test accuracy comparison across the three evaluated architectures.

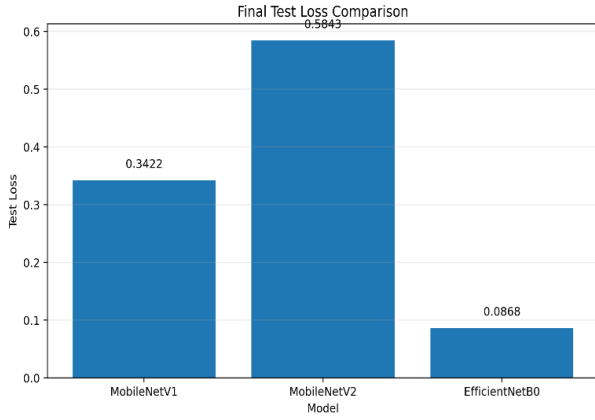


Fig. 9: Final test loss comparison across the three evaluated architectures.

C. Confusion Matrix Analysis

The confusion matrices per class provide additional information in the same vein as the summary statistics. The confusion matrix of EfficientNetB0 is almost diagonal, which implies that most of the classes have been classified correctly. The few non-zero elements outside of the main diagonal, indicating misclassification, are very sparse. In particular, the confusion of some classes, such as Loose Silky-bent and Black-grass, is noteworthy because of their resemblance based on narrow elongated leaves that often confuse other classifiers. MobileNetV1 and MobileNetV2 exhibit groups of confusions between visually similar classes. However, MobileNetV2 shows consistent tendencies towards certain classes being classified as belonging to nearby classes. These observations imply that the feature representation learned by MobileNets is suitable for generic image classification but is not capable of extracting differences in leaf morphology and venation.

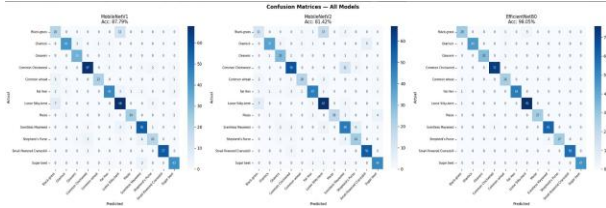


Fig. 10: Confusion matrices for MobileNetV1 (87.79%), MobileNetV2 (81.42%), and EfficientNetB0 (98.05%).

D. Discussion

In summary, both outcomes support two general conclusions. The first is that transfer learning is a viable approach for image classification in agriculture; even the worst-performing model in this evaluation (MobileNetV2, 81.42%) is well above what one would expect from its training counterpart on this scale. Second, the architecture itself can make a difference beyond sheer number of parameters; the compound scaling used by EfficientNetB0 enables capturing discriminatory characteristics not accessible to the depth wise-separable MobileNet class of models.

In practical terms, for precision agriculture, this means that systems based on EfficientNetB0 are fit for implementing selective herbicide spraying, constant weed pressure tracking, and risk of crop yield analysis with enough precision to be usable in a real-world setting. Models using the MobileNetV1 architecture could still be used for scouting operations where classification is less important than categorization, provided user expectations about their error rate are appropriately managed.

V. FINAL PERFORMANCE SUMMARY

Table II summarizes the results on test data for the three models.

TABLE II
FINAL TEST PERFORMANCE OF THE EVALUATED MODELS

Model	Test Accuracy (%)	Test Loss
MobileNetV1	87.79	0.3422
MobileNetV2	81.42	0.5843
EfficientNetB0	98.05	0.0868

VI. CONCLUSION

The main aim of this study was to determine which pre-trained architecture, be it MobileNetV1, MobileNetV2, or EfficientNetB0, could give the best results in segregating weeds from crop plants when fine-tuned on the twelve-class V2 Plant Seedling Dataset through the same consistent two-stage process of training. EfficientNetB0 clearly triumphs in the contest with 98.05% compared to the two contenders.

The accuracy of MobileNetV1 is acceptable at 87.79% of MobileNetV2 at 81.42% be wrong with the inverted residual bottleneck design hindering texture analysis crucial for distinguishing seedlings. Apart from the figures, it is clear that transfer learning could provide a practical approach to automate weed recognition in agricultural settings that lack sufficient resources. This conclusion came from the successful application of the two-stage approach with two stages of feature extraction and fine-tuning being implemented.

Directions for future research can be taken from several angles. For instance, evaluating how well the more sophisticated models such as EfficientNetB4 or ResNet-50 can perform on larger and diversified datasets will help establish the reliability of the findings. Using the best-performing model for an inference pipeline for the embedded hardware and assigning coordinates to weed clusters can convert this research project into its application version.

REFERENCES

- [1] J. A. Jose, A. Sharma, M. Sebastian, and F. D. Raj V, "Classification of weeds and crops using transfer learning," in Proc. Int. Conf. Advances in Computing, Communication and Applied Informatics (ACCAI), 2022, doi: 10.1109/ACCAI53970.2022.9752477.
- [2] D. Vypirailenko, E. Kiseleva, D. Shadrin, and M. Pukalchik, "Deep learning techniques for enhancement of weeds growth clas-sification," in Proc. IEEE Int. Instrumentation and Measurement Technology Conf. (I2MTC), Glasgow, UK, 2021, pp. 1–6, doi: 10.1109/I2MTC50364.2021.9459976.
- [3] B. Espejo-Garcia, N. Mylonas, L. Athanasakos, S. Fountas, and I. Vasilakoglou, "Towards weeds identification assistance through trans-fer learning," Computers and Electronics in Agriculture, vol. 171, p. 105306, Apr. 2020, doi: 10.1016/j.compag.2020.105306.
- [4] S. Mujtaba et al., "Revolutionizing agriculture: Machine and deep learning solutions for enhanced crop quality and weed control," IEEE Access, vol. 12, 2024, doi: 10.1109/ACCESS.2024.3355017.
- [5] A. S. M. M. Hasan, F. Sohel, D. Diepeveen, H. Laga, and M. G. K. Jones, "A survey of deep learning techniques for weed detection from images," Computers and Electronics in Agriculture, vol. 184, p. 106067, May 2021, doi: 10.1016/j.compag.2021.106067.
- [6] A. Wang, W. Zhang, and X. Wei, "A review on weed detection using ground-based machine vision and image processing techniques," Computers and Electronics in Agriculture, vol. 158, pp. 226–240, Mar. 2019, doi: 10.1016/j.compag.2019.02.005.
- [7] M. Bah, A. Hafiane, and R. Canals, "Deep learning with unsuper-vised data labeling for weed detection in line crops," Computers and Electronics in Agriculture, vol. 163, p. 104840, Aug. 2019, doi: 10.1016/j.compag.2019.104840.
- [8] A. Kamilaris and F. X. Prenafeta-Boldu', "Deep learning in agriculture: A survey," Computers and Electronics in Agriculture, vol. 147, pp. 70–90, Apr. 2018, doi: 10.1016/j.compag.2018.02.016.
- [9] S. Sladojevic, M. Arsenovic, A. Anderla, D. Culibrk, and D. Stefanovic, "Deep neural networks based recognition of plant diseases by leaf image classification," Computational Intelligence and Neuroscience, vol. 2016, doi: 10.1155/2016/3289801.
- [10] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classifica-tion with deep convolutional neural networks," in Advances in Neural Information Processing Systems (NeurIPS), 2012.
- [11] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolu-tional neural networks," in Proc. Int. Conf. Machine Learning (ICML), vol. 97, pp. 6105–6114, 2019.
- [12] A. G. Howard et al., "MobileNets: Efficient convolutional neural net-works for mobile vision applications," arXiv:1704.04861, 2017.
- [13] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2018.
- [14] J. Deng et al., "ImageNet: A large-scale hierarchical image database," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2009.
- [15] J. G. A. Barbedo, "A review on the use of computer vision and artificial intelligence in weed control," Computers and Electronics in Agriculture, vol. 153, pp. 470–485, Oct. 2018, doi: 10.1016/j.compag.2018.08.031.